



9305 - Progenitor Properties of the Magnetar-Hosting SNR Kes 73

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	MIRI imaging mosaic	MIRI Imaging	(1) Kes73
Observation Folder				
	2	MRS-position1	MIRI Medium Resolution Spectroscopy	(2) Kes73-position1
	3	MRS-position2	MIRI Medium Resolution Spectroscopy	(3) Kes73-position2
	4	MRS-background	MIRI Medium Resolution Spectroscopy	(4) Kes73-background

ABSTRACT

We propose Chandra and JWST observations of the Galactic magnetar-hosting supernova remnant Kes 73 with the goal of constraining the properties of its progenitor star. Previous X-ray studies of this SNR measured enhanced ejecta abundances consistent with a low progenitor mass of <15 solar masses, a result with significant implications for understanding magnetar formation. However, there is evidence that unaccounted-for line-of-sight emission from shocked ambient medium in Kes 73 may bias results towards lower mass progenitors. The joint X-ray and IR analysis will allow us to disentangle the SN ejecta and swept-up CSM components and characterize their mass, chemical abundances, and dust properties. The study will

constrain the mass of the progenitor and its pre-SN mass loss. Chandra proposal number: 27500281

OBSERVING DESCRIPTION

We request 3×5 imaging mosaics with the JWST MIRI instrument covering the entire SNR, and MIRI MRS spectroscopy in the 5- 28 m wavelength range for two select positions, one centered on an ejecta knot and the other in a region dominated by CSM. Both imaging and MRS observations employ a four-point extended source dither pattern. The MRS observations include a dedicated off-source background pointing. We have linked the MRS background observation as a non-interruptible sequence with the on-source MRS observations. We also imposed a requirement for the MRS observations to be executed after the imaging observations and places the MRS observations on hold until the precise IFU coordinates have been verified based on the MIRI imaging mosaic.

Proposal 9305 - Targets - Progenitor Properties of the Magnetar-Hosting SNR Kes 73

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	Kes73	RA: 18 41 19.9313 (280.3330471d) Dec: -04 56 15.67 (-4.93769d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Category=ISM</i> <i>Description=[Supernova remnants]</i> <i>Extended=YES</i>	Epoch of Position: 2000	
	(2)	Kes73-position1	RA: 18 41 17.2269 (280.3217788d) Dec: -04 57 19.99 (-4.95555d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Category=ISM</i> <i>Description=[Supernova remnants]</i> <i>Extended=YES</i>	Epoch of Position: 2000	
	(3)	Kes73-position2	RA: 18 41 17.8823 (280.3245096d) Dec: -04 56 40.17 (-4.94449d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Category=ISM</i> <i>Description=[Supernova remnants]</i> <i>Extended=YES</i>	Epoch of Position: 2000	
	(4)	Kes73-background	RA: 18 41 24.8059 (280.3533579d) Dec: -04 52 40.06 (-4.87779d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Category=ISM</i> <i>Description=[Supernova remnants]</i> <i>Extended=YES</i>	Epoch of Position: 2000	

Proposal 9305 - Observation 1 - Progenitor Properties of the Magnetar-Hosting SNR Kes 73

Observation	Proposal 9305, Observation 1: MIRI imaging mosaic										Wed Oct 29 17:00:53 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:7) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:8) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:9) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:10) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	Kes73	RA: 18 41 19.9313 (280.3330471d) Dec: -04 56 15.67 (-4.93769d) Equinox: J2000			Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=ISM Description=[Supernova remnants] Extended=YES										
Template	Subarray										
	FULL										
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order
	3	5	10.0		10.0		0.0		0.0		DEFAULT
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4		6	1			LARGE	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F770W	FASTR1	25	1	1	Dither 1	4	4	277.504	
	2	F1130W	FASTR1	25	1	1	Dither 1	4	4	277.504	
	3	F1280W	FASTR1	25	1	1	Dither 1	4	4	277.504	
	4	F1500W	FASTR1	20	1	1	Dither 1	4	4	222.003	
	5	F1800W	FASTR1	15	2	1	Dither 1	4	8	344.105	
	6	F2100W	FASTR1	10	3	1	Dither 1	4	12	355.205	
	7	F2550W	FASTR1	10	3	1	Dither 1	4	12	355.205	

Proposal 9305 - Observation 1 - Progenitor Properties of the Magnetar-Hosting SNR Kes 73

Special Requirements	Sequence Visits within 53.0 Days Visits Same PA 2 After 1
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Proposal 9305 - Observation 2 - Progenitor Properties of the Magnetar-Hosting SNR Kes 73

Observation	Proposal 9305, Observation 2: MRS-position1												Wed Oct 29 17:00:53 GMT 2025
	Diagnostic Status: Warning												
Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(2)	Kes73-position1	RA: 18 41 17.2269 (280.3217788d)					Epoch of Position: 2000					
			Dec: -04 57 19.99 (-4.95555d)										
			Equinox: J2000										
			Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=ISM Description=[Supernova remnants] Extended=YES										
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
		All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					EXTENDED SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	FASTR1	50	1	1	Dither 1	4	4	555.008	
	1	SHORT(A)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
	1	SHORT(A)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	
	2		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008	
	2	MEDIUM(B)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	
	3		IMAGER	F1000W	FASTR1	50	1	1	Dither 1	4	4	555.008	
	3	LONG(C)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
	3	LONG(C)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	

Proposal 9305 - Observation 2 - Progenitor Properties of the Magnetar-Hosting SNR Kes 73

Special Requirements	On Hold Observations on hold until precise IFU coordinates are selected based on MIRI imaging. 2 After 1 Sequence Observations 2, 3, 4, Non-interruptible
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Proposal 9305 - Observation 3 - Progenitor Properties of the Magnetar-Hosting SNR Kes 73

Observation	Proposal 9305, Observation 3: MRS-position2												Wed Oct 29 17:00:53 GMT 2025	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous				
	(3)	Kes73-position2		RA: 18 41 17.8823 (280.3245096d) Dec: -04 56 40.17 (-4.94449d) Equinox: J2000			Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	Category=ISM													
	Description=[Supernova remnants]													
Extended=YES														
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction			
		All MRS			YES			FULL			Allow Auto Reorder			
Dithers	#	Dither Type					Optimized For			Direction				
	1	4-Point					EXTENDED SOURCE			NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F560W	FASTR1	50	1	1	Dither 1	4	4	555.008		
	1	SHORT(A)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	1	SHORT(A)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		
	2		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008		
	2	MEDIUM(B)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		
	3		IMAGER	F1000W	FASTR1	50	1	1	Dither 1	4	4	555.008		
	3	LONG(C)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	3	LONG(C)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		

Special Requirements	Sequence Observations 2, 3, 4, Non-interruptible
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Proposal 9305 - Observation 4 - Progenitor Properties of the Magnetar-Hosting SNR Kes 73

Observation	Proposal 9305, Observation 4: MRS-background												Wed Oct 29 17:00:53 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(4)	Kes73-background	RA: 18 41 24.8059 (280.3533579d) Dec: -04 52 40.06 (-4.87779d) Equinox: J2000 Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=ISM Description=[Supernova remnants] Extended=YES					Epoch of Position: 2000					
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
		All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					EXTENDED SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	FASTR1	50	1	1	Dither 1	4	4	555.008	
	1	SHORT(A)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
	1	SHORT(A)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	
	2		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008	
	2	MEDIUM(B)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	
	3		IMAGER	F1000W	FASTR1	50	1	1	Dither 1	4	4	555.008	
	3	LONG(C)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
	3	LONG(C)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	

Special Requirements	Sequence Observations 2, 3, 4, Non-interruptible
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